

The Impact of AI-Enhanced YouTube Features on Undergraduate Academic Outcomes

Rajendran Adshayan

Department of Computer Science, Faculty of Science, University of Jaffna

Jeevathas Ushalini

Department of Information Technology, Faculty of Management Studies and Commerce, University of Sri Jayewardenepura

ICODE2026

03-Aug-26

UNIVERSITY OF MORATUWA

Centre for Open and Distance Learning

Introduction

Background: YouTube launched in 2005 as an entertainment platform, but now hosts vast academic content — lectures, tutorials, and peer-produced explanations that enhance learning.

Adoption: Widely used as a supplementary learning platform in higher education due to its accessibility and visual format, a trend mirrored by Sri Lankan university students.

Role of AI: AI simulates human intelligence — learning, reasoning, self-correction — manifesting on YouTube as auto-captions, personalized recommendations, and content curation.

Impact: These features analyze viewer behavior and learning patterns to enhance engagement, providing students with quicker access to relevant content and improved comprehension.

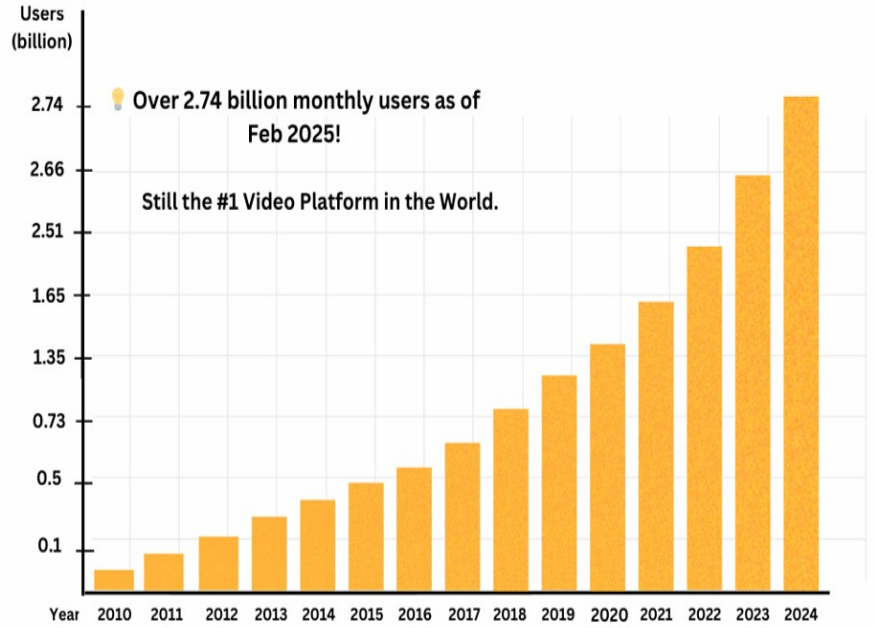


Figure 1: YouTube's Global Growth 2010–2024

Problem Identification

- YouTube usage for academic learning is high among Sri Lankan undergraduates, yet its AI-driven features remain largely unexamined for their actual academic outcome.
- AI-enhanced features such as recommendations, subtitles, and curated content may support or hinder academic performance, but their actual effect remains unclear.
- There is a lack of comprehensive studies examining how AI-enhanced YouTube features specifically influence academic outcomes within the Sri Lankan context.
- This research problem centers on the influence of AI-enhanced YouTube features on the academic outcomes of undergraduate students.

Research Questions

- **RQ1:** What is the relationship between the frequency of using YouTube's AI features and the academic outcomes of undergraduate students?
- **RQ2:** Which YouTube AI feature has the most significant impact on the academic performance of undergraduate students?
- **RQ3:** How do undergraduate students perceive the usefulness of YouTube's AI features in improving academic understanding and learning efficiency?

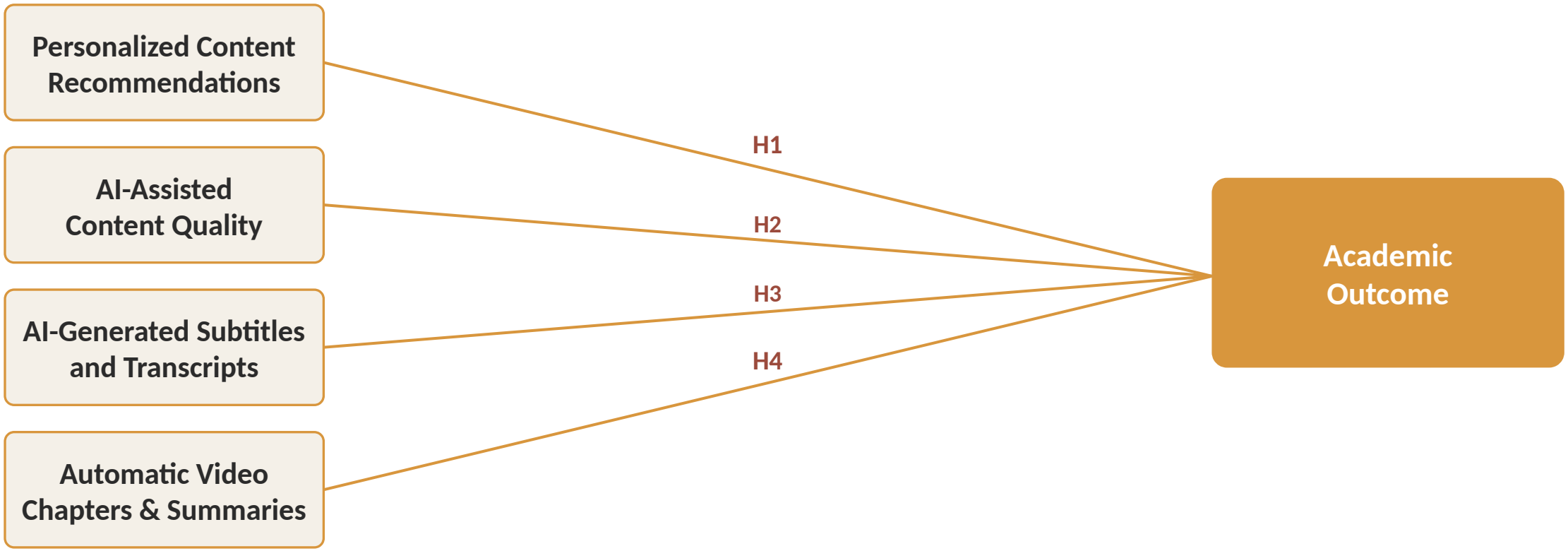
Research Objectives

- **RO1**: To examine the relationship between the frequency of using YouTube AI features and the academic outcomes of undergraduate students.
- **RO2**: To identify which YouTube AI feature has the most significant impact on the academic performance of undergraduate students.
- **RO3**: To assess undergraduate students' perceptions of the usefulness of YouTube AI features in improving academic understanding and learning efficiency.

Literature Review

- **Information Adoption Model:** Perceived credibility, argument quality, and usefulness drive information adoption; learners more readily adopt AI-curated, relevant videos.
- **Media Richness Theory:** A medium's richness — feedback immediacy, language variety, personalization — shapes learning value; supports AI subtitles and video chapters.
- **Cognitive Load Theory:** Instructional design affects working memory; segmenting videos into chapters enables incremental, focused learning.
- **Connectivism & Self-Directed Learning:** Knowledge is distributed across networks; AI recommendations and playlists support personalized, self-paced learning and learner ownership.

Methodology: Conceptual Framework



Hypothesis Development

- **H1:** Personalized Content Recommendations have a significant influence on the academic outcomes of undergraduate students using YouTube for learning.
- **H2:** AI-Assisted Content Quality has a significant influence on the academic outcomes of undergraduate students using YouTube for learning.
- **H3:** AI-Generated Subtitles and Transcripts have a significant influence on the academic outcomes of undergraduate students using YouTube for learning.
- **H4:** Automatic Video Chapters and Summaries have a significant influence on the academic outcomes of undergraduate students using YouTube for learning.

Data Analysis

- **Approach:** Quantitative research approach using a structured online questionnaire.
- **Population:** Undergraduate students at the Faculty of Management Studies and Commerce, University of Sri Jayewardenepura with prior experience using YouTube for academic learning.
- **Required Sample Size:** Minimum of 122, based on Cohen's guidelines at 5% significance level with minimum R^2 of 0.10
- **Analysis Technique:** Partial Least Squares Structural Equation Modeling (PLS-SEM), analysed using SmartPLS 4 and IBM SPSS Statistics.

Data Collection & Measurement Model

- **Model Type:** 125 valid responses analyzed using a reflective measurement model for five constructs: PCR, AACQ, AST, AVCS, and Academic Outcome (AO)
- **Internal Consistency Reliability:** Assessed via Cronbach's Alpha and Composite Reliability — all constructs met acceptable thresholds
- **Convergent Validity:** Assessed via outer loadings and Average Variance Extracted (AVE)
- **Discriminant Validity:** Assessed via cross-loadings, Fornell–Larcker criterion, and HTMT ratio — confirming all constructs are distinct
- **Outcome:** Measurement model satisfied all reliability and validity criteria, confirming suitability for structural model analysis

Assessment of Structural Model

Hyp.	Path	Coefficient	t-value	p-value	Result
H1	PCR → AO	0.241	3.865	0.005	Supported
H2	AACQ → AO	0.098	1.645	0.100	Not Supported
H3	AST → AO	0.112	1.732	0.084	Not Supported
H4	AVCS → AO	0.318	4.210	0.001	Supported

Significance thresholds: t-value > 1.96, p-value < 0.05.

H1 (Personalized Recommendations) and H4 (Automatic Video Chapters & Summaries) are supported.

H2 (AI-Assisted Content Quality) and H3 (AI-Generated Subtitles/Transcripts) are not supported.

Conclusion

- **Key Finding:** Personalized Content Recommendations and Automatic Video Chapters & Summaries significantly improve undergraduate academic outcomes; AI-Assisted Content Quality and AI-Generated Subtitles/Transcripts show no significant effect
- **Implication:** AI features that support content discovery and structured navigation add more academic value than quality or accessibility features
- **Limitation:** Academic outcomes were measured using self-reported GPA and satisfaction, which may not fully capture actual performance
- **Future Research:** Future studies should incorporate objective academic records and longitudinal designs for a more comprehensive evaluation
- **Practical Value:** Educators and platforms should prioritize personalization and structured content navigation when designing AI-enabled learning tools

Thank you for your attention.